

PSET 6: Discrete random variables

Problem 1: Calculate probabilities in a sample space S

Events A and B are contained within a sample space S . Given that $\Pr(A) = 0.55$, $\Pr(B) = 0.3$, and $\Pr(A \cap B) = 0.15$, find:¹

- $\Pr(A \cup B)$
- $\Pr(A \cap B^c)$
- $\Pr[(A \cap B^c) \cup (B \cap A^c)]$
- Are A and B independent? Are they mutually exclusive? Explain how these concepts differ.

Answer

- $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B) = 0.55 + 0.3 - 0.15 = 0.70$.
- $\Pr(A \cap B^c) = \Pr(A) - \Pr(A \cap B) = 0.55 - 0.15 = 0.40$.
- $\Pr[(A \cap B^c) \cup (B \cap A^c)] = \Pr(A \cap B^c) + \Pr(B \cap A^c) = 0.40 + 0.15 = 0.55$. (Equivalently, this is $\Pr(A \cup B) - \Pr(A \cap B) = 0.70 - 0.15 = 0.55$.)
- Independence:** A and B are independent if and only if $\Pr(A \cap B) = \Pr(A)\Pr(B)$. Here $\Pr(A)\Pr(B) = 0.55 \times 0.3 = 0.165$, but $\Pr(A \cap B) = 0.15 \neq 0.165$, so A and B are not independent.

Mutual exclusivity: A and B are not mutually exclusive, because $\Pr(A \cap B) = 0.15 \neq 0$ — it is possible for both to occur at once.

How the concepts differ: Mutual exclusivity asks whether two events can occur simultaneously. Independence asks whether knowing one event occurred changes the probability of the other. In general, if two events are mutually exclusive and both have positive probability, they will not be independent. This is because if A occurs, B is guaranteed not to occur — so learning that A happened completely changes what we know about B . That is the opposite of independence, which requires that learning about A tells us nothing new about B .

Problem 2: Random variables

Provide the following explanations in your own words:

- What is a random variable?
- What is the difference between upper case and lower case x ? How (if at all) do they matter?

¹Inspired by Grimmer HW8.4

- c. If I'm thinking about something like $p_X(x_0)$, what do all the parts/pieces mean?
- d. What is a discrete random variable?

Answer

- a. A random variable is a function that assigns a real number to every outcome in a sample space. Formally, if S is the sample space, a random variable is a function $X : S \rightarrow \mathbb{R}$.
- b. Upper case X denotes the assigning function itself, whereas lower case x denotes a single fixed value that the function might produce. This separation is useful because it allows us to evaluate the likelihood of particular results, such as in $p(X = x)$, which asks for the probability that the function's output equals x .
- c. p_X is the probability mass function belonging to the random variable X . x_0 is a particular value in the support of X . So $p_X(x_0) = \Pr(X = x_0)$ is the probability that X takes on the specific value x_0 .
- d. A discrete random variable is a random variable whose possible values form a countable set — either finite (e.g. $\{0, 1, 2, 3\}$) or countably infinite (e.g. $\{0, 1, 2, \dots\}$).

Problem 3: Survey says

A survey has 54% respondents 50 or older and 46% respondents under 50. Within the survey, on a particular question, 9.5% of the 50-plus population agrees strongly while 2.7% of under-50 respondents agree strongly.

- a. What is the probability someone selected at random is 50 or older?
- b. The selected individual strongly agrees with the survey question. Now what is the likelihood that person is 50 or older? Explain your reasoning and show all your work.
- c. Are the two answers above the same or different? Explain.
- d. (for fun, no points) What is the survey question?

Answer

Let O = “respondent is 50 or older” and A = “respondent strongly agrees.” We're given $\Pr(O) = 0.54$, $\Pr(O^c) = 0.46$, $\Pr(A | O) = 0.095$, and $\Pr(A | O^c) = 0.027$.

- a. $\Pr(O) = 0.54$.
- b. First find $\Pr(A)$ using the law of total probability:

$$\Pr(A) = \Pr(A | O) \Pr(O) + \Pr(A | O^c) \Pr(O^c) = (0.095)(0.54) + (0.027)(0.46) = 0.0513 + 0.01242 = 0.06372.$$

Then by Bayes' rule,

$$\Pr(O | A) = \frac{\Pr(A | O) \Pr(O)}{\Pr(A)} = \frac{0.0513}{0.06372} \approx 0.805.$$

So given that the person strongly agrees, there's about an 80.5% chance they are 50 or older.

- c. The two answers are different. Part (a) is the unconditional probability of being 50 or older. Part (b) is a conditional probability that incorporates additional information: once we learn the person strongly agreed, this makes it more likely they are in the 50-plus group, because strong agreement is roughly 3.5 times more common among the 50-plus group (9.5%) than the under-50 group (2.7%).
- d. E.g. a question about Social Security, mandatory retirement savings, or a similar policy that older respondents would be expected to favor more strongly than younger ones.

Problem 4: PMF vs. CMF

Consider the following function: $f(x) = \frac{1}{6}$.

- a. Find the pmf and cmf of the function and provide them in a table below. *You may need to make some assumptions for this to work – clearly state any assumptions you make*
- b. What is the variance of X ?
- c. Now consider a different function, $p(x) = cx$ for $x \in \{1, 2, 3, 4\}$ and 0 otherwise. For what value of c is this a valid pmf? State which property of a pmf you used to find it.
- d. Using the value of c you found, calculate $E[X]$ and $\text{Var}(X)$.

Answer

- a. **Assumption:** for $f(x) = 1/6$ to be a valid pmf, it needs a finite support over which the probabilities sum to 1. Six equally likely outcomes do that, so assume X is uniform on $x \in \{1, 2, 3, 4, 5, 6\}$ (e.g. a fair six-sided die), and $f(x) = 0$ otherwise. Then $p_X(x) = 1/6$ for each x , and the cmf is the running sum:

x	1	2	3	4	5	6
$p_X(x)$	1/6	1/6	1/6	1/6	1/6	1/6
$F_X(x)$	1/6	2/6	3/6	4/6	5/6	1

b.

$$E[X] = \sum_{x=1}^6 x \cdot \frac{1}{6} = \frac{21}{6} = 3.5, \quad E[X^2] = \sum_{x=1}^6 x^2 \cdot \frac{1}{6} = \frac{91}{6} \approx 15.167.$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{91}{6} - 3.5^2 = 15.167 - 12.25 = \frac{35}{12} \approx 2.917.$$

- c. A valid pmf must sum to 1 over its support. So we need

$$\sum_{x=1}^4 cx = c(1 + 2 + 3 + 4) = 10c = 1 \implies c = \frac{1}{10}.$$

- d. With $c = 1/10$, $p(x) = x/10$ for $x \in \{1, 2, 3, 4\}$.

$$E[X] = \sum_{x=1}^4 x \cdot \frac{x}{10} = \frac{1 + 4 + 9 + 16}{10} = \frac{30}{10} = 3.$$

$$E[X^2] = \sum_{x=1}^4 x^2 \cdot \frac{x}{10} = \frac{1 + 8 + 27 + 64}{10} = \frac{100}{10} = 10.$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 10 - 3^2 = 1.$$

Problem 5: Building a distribution from scratch

Three countries are independently deciding whether to sign a treaty. Each country signs with probability 0.6. Let X be the number of countries that sign.

- Write out the sample space. How many outcomes does it contain?
- Construct the pmf of X . Show how you collected outcomes together to obtain $p_X(x)$ for each value of x .
- Verify that your pmf satisfies the properties of a valid pmf.
- Construct the cmf of X and present it in a table alongside the pmf.
- Find $E[X]$ and $\text{Var}(X)$.
- Is X a named distribution? If so, which one, and with what parameters? Do your answers to (e) match the formulas for that distribution's mean and variance?

Answer

- Label each country's outcome S (signs) or N (does not sign). Since there are 3 countries each with 2 possible outcomes, the sample space is

$$\Omega = \{SSS, SSN, SNS, NSS, SNN, NSN, NNS, NNN\},$$

which contains $2^3 = 8$ outcomes.

- X counts the number of S's in the outcome. Group the outcomes by how many countries signed:
 - $X = 0$: $\{NNN\}$ (1 outcome), $p_X(0) = (0.4)^3 = 0.064$.
 - $X = 1$: $\{SNN, NSN, NNS\}$ (3 outcomes), each with probability $0.6 \times 0.4^2 = 0.096$, so $p_X(1) = 3 \times 0.096 = 0.288$.
 - $X = 2$: $\{SSN, SNS, NSS\}$ (3 outcomes), each with probability $0.6^2 \times 0.4 = 0.144$, so $p_X(2) = 3 \times 0.144 = 0.432$.
 - $X = 3$: $\{SSS\}$ (1 outcome), $p_X(3) = (0.6)^3 = 0.216$.
- A valid pmf requires (i) $p_X(x) \geq 0$ for all x , and (ii) $\sum_x p_X(x) = 1$. Both hold here: each probability computed in part (b) is positive, and

$$0.064 + 0.288 + 0.432 + 0.216 = 1.000. \checkmark$$

	x	0	1	2	3
d.	$p_X(x)$	0.064	0.288	0.432	0.216
	$F_X(x)$	0.064	0.352	0.784	1.000

e.

$$E[X] = \sum_x x p_X(x) = 0(0.064) + 1(0.288) + 2(0.432) + 3(0.216) = 1.8.$$

$$E[X^2] = 0^2(0.064) + 1^2(0.288) + 2^2(0.432) + 3^2(0.216) = 0.288 + 1.728 + 1.944 = 3.96.$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 3.96 - 1.8^2 = 3.96 - 3.24 = 0.72.$$

f. Yes: $X \sim \text{Binomial}(n = 3, p = 0.6)$, since X counts the number of successes in 3 independent trials each with success probability 0.6. The binomial formulas give

$$E[X] = np = 3(0.6) = 1.8, \quad \text{Var}(X) = np(1 - p) = 3(0.6)(0.4) = 0.72,$$

which match the values computed directly in (e).

Problem 6: Getting a traffic ticket

You drive to work 5 days a week for 10 weeks, and with probability $p = 0.06$ you get a traffic ticket on any given day, independent of other days. Let X be the total number of tickets you get in this time.²

- What is the probability that the number of tickets you get is exactly equal to the expected value of X ?
- Calculate approximately the probability in (a) using a Poisson approximation.
- Now suppose you are instead interested in how long you can go before your first ticket. What is the probability that your first ticket comes on the 10th day you drive?
- What is the expected number of days you drive until you get your first ticket? Which distribution did you use here, and why is it a different one from the distribution you used in (a)?

Answer

There are $n = 5 \times 10 = 50$ driving days total, and $X \sim \text{Binomial}(n = 50, p = 0.06)$.

- First find the expected value:

$$E[X] = np = 50(0.06) = 3.$$

So we want $\Pr(X = 3)$. Using the binomial pmf,

$$\Pr(X = 3) = \binom{50}{3} (0.06)^3 (0.94)^{47}.$$

Computing each piece:

$$\binom{50}{3} = \frac{50 \times 49 \times 48}{3 \times 2 \times 1} = 19,600, \quad (0.06)^3 = 0.000216, \quad (0.94)^{47} \approx 0.05455.$$

Multiplying these together:

$$\Pr(X = 3) \approx 19,600 \times 0.000216 \times 0.05455 = 4.2336 \times 0.05455 \approx 0.2310.$$

²Inspired by BT 2.41

- b. Using the Poisson approximation with $\lambda = np = 3$:

$$\Pr(X = 3) \approx e^{-3} \frac{3^3}{3!} = e^{-3}(4.5) \approx 0.2240$$

- c. We want the probability that the first 9 days have no ticket and day 10 does:

$$\Pr(\text{first ticket on day 10}) = (1 - p)^9 p = (0.94)^9 (0.06) \approx 0.0344.$$

- d. If Y is the number of days until (and including) the first ticket, $Y \sim \text{Geometric}(p = 0.06)$. The expected value of a geometric random variable is $E[Y] = 1/p$, so

$$E[Y] = \frac{1}{0.06} \approx 16.67 \text{ days.}$$

This uses the geometric distribution rather than the binomial from (a) because the binomial in (a) counts the number of successes in a fixed number of trials (50 days), whereas here the number of trials itself is random: we keep driving until the first ticket occurs, and we are asking how long that takes on average.

Problem 7: Obtaining requests for information

X is a discrete random variable. It takes the value of the number of days required for a governmental agency to respond to a request for information. X is distributed according to the following PMF:³

$$f(x) = e^{-6} \frac{6^x}{x!} \quad \text{for } x \in \{0, 1, 2, \dots\}$$

- Given this information, what is the probability of a response from the agency in 5 days or less?
- What is the probability the agency response takes more than 10 but less than 13 days?
- What is the probability the agency response takes more than 5 days?
- Suppose using X you generate a new variable, **Responsive**. **Responsive** equals 1 if an agency responds in 5 days or less and 0 otherwise. What is the expected value of **Responsive**?
- What is the variance of **Responsive**?
- The agency charges a processing fee for each request: a flat \$25.00, plus \$10.00 for every day it takes them to respond. Let $C = 25 + 10X$ be the fee you are charged. Find $E[C]$ and $\text{Var}(C)$. **Hint: you should not need to recompute anything from scratch.**

Answer

$X \sim \text{Poisson}(\lambda = 6)$, so $e^{-6} \approx 0.0024788$.

³Inspired by Grimmer HW10.2

- a. We need $\Pr(X \leq 5) = \sum_{x=0}^5 e^{-6} \frac{6^x}{x!}$. Computing each term:

$$\begin{aligned} p(0) &= e^{-6} \frac{6^0}{0!} = e^{-6}(1) \approx 0.002479, \\ p(1) &= e^{-6} \frac{6^1}{1!} = e^{-6}(6) \approx 0.014873, \\ p(2) &= e^{-6} \frac{6^2}{2!} = e^{-6}(18) \approx 0.044619, \\ p(3) &= e^{-6} \frac{6^3}{3!} = e^{-6}(36) \approx 0.089235, \\ p(4) &= e^{-6} \frac{6^4}{4!} = e^{-6}(54) \approx 0.133853, \\ p(5) &= e^{-6} \frac{6^5}{5!} = e^{-6}(64.8) \approx 0.160623. \end{aligned}$$

Adding these up:

$$\Pr(X \leq 5) \approx 0.002479 + 0.014873 + 0.044619 + 0.089235 + 0.133853 + 0.160623 \approx 0.4457.$$

- b. We need $p(11)$ and $p(12)$ individually.

$$\begin{aligned} p(11) &= e^{-6} \frac{6^{11}}{11!} = e^{-6} \frac{362,797,056}{39,916,800} \approx e^{-6}(9.0902) \approx 0.0225. \\ p(12) &= e^{-6} \frac{6^{12}}{12!} = e^{-6} \frac{2,176,782,336}{479,001,600} \approx e^{-6}(4.5451) \approx 0.0113. \end{aligned}$$

So

$$\Pr(10 < X < 13) = p(11) + p(12) \approx 0.0225 + 0.0113 = 0.0338.$$

- c. $\Pr(X > 5) = 1 - \Pr(X \leq 5) = 1 - 0.4457 = 0.5543$.
d. **Responsive** is a Bernoulli random variable, since it only takes values 0 or 1, with

$$\Pr(\text{Responsive} = 1) = \Pr(X \leq 5).$$

For a Bernoulli random variable, $E[\text{Responsive}] = 1 \cdot \Pr(\text{Responsive} = 1) + 0 \cdot \Pr(\text{Responsive} = 0) = \Pr(\text{Responsive} = 1)$. Using the value from (a):

$$E[\text{Responsive}] = \Pr(X \leq 5) \approx 0.4457.$$

- e. For a Bernoulli random variable with success probability π ,

$$\text{Var}(\text{Responsive}) = E[\text{Responsive}^2] - (E[\text{Responsive}])^2 = \pi - \pi^2 = \pi(1 - \pi),$$

where the first equality uses $\text{Responsive}^2 = \text{Responsive}$ (since it only takes values 0 and 1). Here $\pi = 0.4457$ from part (d), so

$$\text{Var}(\text{Responsive}) = 0.4457(1 - 0.4457) = 0.4457 \times 0.5543 \approx 0.2470.$$

- f. Let $C = g(X) = 25 + 10X$. Using the expected value rule for functions of a random variable, $\mathbb{E}[g(X)] = \sum_x g(x) p_X(x)$, and the fact that for a Poisson random variable $\mathbb{E}[X] = \text{Var}(X) = \lambda = 6$:

$$\mathbb{E}[C] = \sum_x (25 + 10x) p_X(x) = 25 \sum_x p_X(x) + 10 \sum_x x p_X(x) = 25 + 10 \mathbb{E}[X] = 25 + 10(6) = 85.$$

For the variance, using $\text{Var}(C) = \mathbb{E}[C^2] - \mathbb{E}[C]^2$: since $C = 25 + 10X$ shifts X by a constant and scales it by 10, only the scaling affects spread around the mean, so

$$\text{Var}(C) = 10^2 \text{Var}(X) = 100(6) = 600.$$

Problem 8: Modeling electoral outcomes

Suppose we've developed a model predicting the outcome of the upcoming midterm elections in a state with 4 Congressional districts. In each district there are two candidates, a Republican and a Democrat. We have reason to believe the following PMF describes the distribution of potential election results, where $K \in \{0, 1, 2, 3, 4\}$ is the number of seats won by Republican candidates in the upcoming election.

$$\Pr(K = k \mid \theta) = \binom{4}{k} \theta^k (1 - \theta)^{4-k}$$

Based on polling information, we think the appropriate value for θ is 0.423.⁴

- How many distinct ways can Republicans win exactly 2 of the 4 seats?
- What's the expected number of seats Republicans will win in the upcoming election?
- Given this PMF, what's the probability that no Republican legislators win in the upcoming election?
- What's the probability that Republican legislators win a majority of the seats in this state?
- A prominent political pundit declares they are certain that Republicans will win a majority of seats in the next election and offers the following bet. If Republicans win a majority of the seats, we must pay the pundit \$15.00. If Republicans fail to win a majority of the seats, we will win \$20.00. Based on our model, should we take this bet? **Hint: think of the betting outcomes as a random variable. Find the expected value of this random variable.**
- Suppose we are offered a second bet with a more complicated structure. In this case we'll receive \$100 if the Republicans win a majority, \$50 if neither party wins a majority, and we'll have to pay \$200 if the Democrats win a majority. Should we take this bet?

Answer

$K \sim \text{Binomial}(n = 4, \theta = 0.423)$. The full pmf:

k	0	1	2	3	4
$\Pr(K = k)$	0.1108	0.3250	0.3574	0.1747	0.0320

⁴Inspired by Grimmer HW10.3. Data from <https://www.realclearpolling.com/polls/state-of-the-union/generic-congressional-vote>

- a. $\binom{4}{2} = 6$ distinct ways.
- b. $E[K] = n\theta = 4(0.423) = 1.692$ seats.
- c. $\Pr(K = 0) = (1 - \theta)^4 = (0.577)^4 \approx 0.1108$.
- d. A majority of 4 seats means winning at least 3, i.e. $K \geq 3$:

$$\Pr(K \geq 3) = \Pr(K = 3) + \Pr(K = 4) \approx 0.1747 + 0.0320 = 0.2067.$$

- e. Let W be our winnings from the bet. $W = -15$ if Republicans win a majority (an event with probability 0.2067 from part (d)), and $W = 20$ otherwise (probability $1 - 0.2067 = 0.7933$). Then

$$E[W] = (-15)(0.2067) + (20)(0.7933).$$

Computing each term: $(-15)(0.2067) = -3.1005$ and $(20)(0.7933) = 15.866$, so

$$E[W] \approx -3.1005 + 15.866 = 12.77.$$

Since $E[W] > 0$, we should take this bet: on average, we expect to gain about \$12.77 per bet under our model.

- f. Now there are three outcomes to track:

- Republicans win a majority ($K \geq 3$): probability ≈ 0.2067 from part (d).
- Neither party wins a majority, i.e. a 2-2 tie ($K = 2$): probability ≈ 0.3575 .
- Democrats win a majority ($K \leq 1$): probability $\Pr(K = 0) + \Pr(K = 1) \approx 0.1108 + 0.3251 = 0.4359$.

Let W be our winnings under this second bet:

$$E[W] = 100(0.2067) + 50(0.3575) - 200(0.4359).$$

Computing each term:

$$100(0.2067) = 20.67, \quad 50(0.3575) = 17.875, \quad 200(0.4359) = 87.18.$$

So

$$E[W] \approx 20.67 + 17.875 - 87.18 = -48.64.$$

Since $E[W] < 0$, we should not take this second bet: on average, we expect to lose about \$48.64 per bet under our model.